CSCI 2916 Lab 1 – Body Mass Index (BMI)

BMI is an inexpensive and easy screening method for weight category—underweight, healthy weight, overweight, and obesity.

BMI does not measure body fat directly, but BMI is moderately correlated with more direct measures of body fat. Furthermore, BMI appears to be as strongly correlated with various metabolic and disease outcome as are these more direct measures of body fatness. (About Adult BMI | Healthy Weight, Nutrition, and Physical Activity | CDC), BMI CALCULATOR INCHES AND POUNDS (bmi-calculator.io)

Simple steps to work out your English BMI:

Multiply your height in inches by itself
Divide your weight in pounds (lbs) by your result in step 1
Multiply the result from step 2 by 703.

Example using formula

For an adult with height of 5ft 11 inches and weight of 155 pounds (lbs). Step one is to convert the height into inches only. There are 12 inches in a foot, so we simply multiply the 5ft by 12 and then add the 11 inches. This gives us a total of 71 inches.
Let's plug those figures into our formula:
BMI = 703 × (155 ÷ (71 × 71))
We do the multiplication inside the brackets first:
BMI = 703 × (155 ÷ 5041)
And then...
BMI = 703 × 0.030747867
This gives us a BMI figure of 21.62.

BMI Categories	
BMI Value	**Category**
BMI of lower than 19	Underweight
BMI of 19 to 25(exclusive)	Healthy
BMI of 25 to 30(exclusive)	Overweight
BMI of 30 and above	Obese

Step 1

1. Create a program that does the following
2. Asks the user for his/her name
3. Asks the user for his/her height in feet and inches
    a. Converts the height into inches
4. Asks the user for his/her weight in pounds.
5. Calculates the BMI (see formula above).
6. Outputs in a nicely formatted way the person's name, their BMI and indicates what category the person falls in (see the BMI Categories table (Underweight, Healthy, Overweight, Obese)

Step 2

Thoroughly test your program by entering in various heights and weights

The following table are inputs you can test your program. Test with 4 more heights and weights

	Testing – very important step in programming			
1	Height	Weight	BMI	Category
2	4 feet 7 inches	115	26.7	Overweight
3	6 feet 2 inches	150	19.3	Healthy
4	5 feet 4 inches	105	18.0	Underweight
5				
6				
7				
8				

Step 3

Repeat the program 5 times

Step 4 (extra credit)

Calculate the BMI using the metric system asking for weight in kilograms and height in meters

Formula: weight (kg) / [height (m)]^2

Calculation: [weight (kg) / height (cm) / height (cm)] x 10,000

CDC - Calculating BMI using the metric system - BMI for Age Training Course - DNPAO

Step 5

Submit .java file to your GitHub repository (not Blackboard)

Sample Run (computer output is in bold, user input is in italic)

This is a Body Mass Index Calculator Program
You will be asked for your name, weight in pounds, and height in feet and inches
****
What is your name?
Bob
What is your weight in pounds?
150
What is your height? Input number of feet
6
What is your height? Input number of inches
2
Hi Bob, your BMI is 19.3
Your BMI puts you in the healthy category

BEFORE YOU START WRITING CODE . . . Think about:

- How will you calculate the height in inches?
- How will you calculate the BMI?
- How will you determine what category the BMI result falls into?
- Use a BMI calculator or calculate in Excel so, when you test your program, you will know it is outputting the correct answer

Guidelines for a good program:

The program works, following the dialog and rules above.

- The code is clear and understandable:
  - Properly indented
  - Representative variable names
  - Blank lines separate logical sections of code
  - Appropriate comments included
  - Preamble documentation is included
  - Review program assignment rubric